

Rahul Medicharla

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TECHNICAL SKILLS

Certifications: AWS Solutions Architect Associate, Certified Kubernetes Application Developer (CKAD)

Languages: Go, Python, Javascript, Java, SQL, C#, C

Frameworks: React, Pytorch, .NET, Node.js, Flask, MongoDB, Ruby on Rails

Developer Tools: Kubernetes, Docker, Jenkins, K6, Gatling, New Relic, Snowflake, Git

EXPERIENCE

Associate Software Engineer

August 2025 – current

Capital One

Mclean, VA

- Developed a production-ready shadow validation service using **AWS Kinesis Streams**, **AWS ECS**, and **AWS DynamoDB** that replays real-time production requests to a candidate release stack, verifying release readiness and increasing release velocity by 54%.
- Developed a high-fidelity performance testing infrastructure for an internal financial core that accelerates release testing by 77%, utilizing **AWS Step Functions**, **AWS Lambdas**, **AWS Quicksight**, **K6**, **Gatling**, and **Snowflake**.

Undergraduate Student Researcher

January 2024 – May 2025

PCVLab, The Ohio State University

Columbus, OH

- Innovated MotivNet, a deep learning model for facial emotion recognition that achieved performance within 10% of state-of-the-art AffectNet FER tasks through **ViT-MAE**-based feature extraction and **Attention**-based classification.
- Conducted guided research in a highly selective undergraduate research distinction program that culminated in a published thesis, utilizing technologies such as **Pytorch**, **Weights & Biases**, and **Pandas**.
- Optimized computation by implementing distributed protocols with **SLURM**, resulting in a 43% reduction in training time.

Software Engineering Intern

June 2024 – August 2024

Capital One

Mclean, VA

- Developed a **Kubernetes**-native controller in **Go** to mitigate cloud risk for 150+ **AWS EKS** clusters by surfacing and annotating **Docker** image vulnerabilities onto pods, accelerating the vulnerability remediation timeline by 50%.
- Designed a novel metric for an enterprise **New Relic** dashboard that identifies the vulnerability footprint of 22,500+ pods by generating vulnerability count histograms bucketed across 5 risk labels and exposing it via a **Prometheus** client.
- Optimized operator performance by at least 15% utilizing **AWS Lambda** for session management and templated operator deployment onto a CI/CD pipeline using **Jenkins** and **Helm**.

Software Engineering Intern

May 2023 – August 2023

WillowTree

Columbus, OH

- Led the creation of an end-to-end MVP for a 24/7 customer support chatbot using the **React** framework, collaborating with a cross-functional team, and following **Agile** methodologies.
- Implemented a semantic search and response feature to enhance the customer experience by cutting down customer service response times by around 30% utilizing large language models, vector embeddings, and an **Azure MySQL** database.
- Deployed a custom back-end web API leveraging **Azure Cloud Services**, **.NET**, and **C#**.

EDUCATION

Georgia Institute of Technology

Masters of Science in Computer Science, ML specialization

Atlanta, Georgia

Spring 2026 - current

The Ohio State University

Bachelor of Science in Computer Science Engineering, Minor in Business

Honors Research Distinction in Civil Engineering

Columbus, OH

Spring 2025

GPA: 3.741/4.0

PROJECTS

K6 Open Source Contributor | Go

August 2025 – current

Kubefs | *Go, Kubernetes, Docker, Helm*

May 2024 – September 2025

- Published a CLI tool to **Homebrew** that automates fullstack application development, testing, & deployment onto **K8** clusters by utilizing **Go**, **Cobra**, **Docker**, and **Helm**, resulting in a 23% reduction in time-to-deploy.
- Integrated support for common frameworks including **Cassandra**, **Redis**, and **NextJS** to reduce operational overhead.

Evolate | Python, Pytorch, Scipy, Pandas, Numpy

May 2023 – August 2023

- Engineered a data structure that autonomously switches between different data structures and search algorithms based on user behavioral patterns, resulting in a 11% performance boost in computation speed.
- Trained a custom **PyTorch Neural Network** to determine the optimal implementations based on behavioral metrics such as insertion and deletion frequencies, search predictions, and search randomness.